Code No.: 24CSPC52 | **R24** | **H.T.No.** | | | R | 0 | | | | | |

CMR INSTITUTE OF TECHNOLOGY: HYDERABAD

UGC AUTONOMOUS

III–B. Tech.I–Semester-I- Mid Term Examinations – SEPTEMBER – 2026

INFORMATION AND CYBER SECURITY

(Common to CSE, CSE(AI&ML), CSE(DS))

[Time: 120 Minutes] **[Max. Marks: 30]**

Note: 1. This question paper contains two parts A and B.
2. Part A is compulsory which carries 5 marks. Answer all questions in Part A.
3. Part B consists of 5 questions. Answer all 5 questions. Each question carries 5 marks and may have sub questions.
4. Illustrate your answers with NEAT sketches wherever necessary.

PART-A **5 x 1M=5M**

S. No.		Question	BTL	CO
1	a	What is CIA Traid?	1	1
	b	Define NIA.	1	1
	c	Define encryption and decryption.	1	2
	d	List the four principal components in OpenSSL.	1	2
	e	List out few of the popular cybercrimes.	1	3

PART-B **5 x 5M=25M**

S. No.	Question	BTL	CO
2	Build the model of a Network Security with neat diagram.	3	1
	OR		
3	Explain operating system vulnerabilities and the possible attacks associated with them.	2	1
4	Explain Security Mechanisms in Information and Cyber Security.	2	1
	OR		
5	Explain the role of an Incident Response Team.	2	1
6	Explain Digital Signature Algorithm with neat diagram.	2	2
	OR		
7	Illustrate with a neat diagram the different phases in a SSL Architecture.	2	2
8	Explain IP Sec (IP Security) Protocols with neat Diagram.	2	2
	OR		
9	Explain the SHA-512 algorithm in detail.	2	2
10	Outline how cybercriminals plan an attack?	2	3
	OR		
11	Define cyber offences and cybercrime. Discuss in detail Botnets with necessary diagram.	2	3